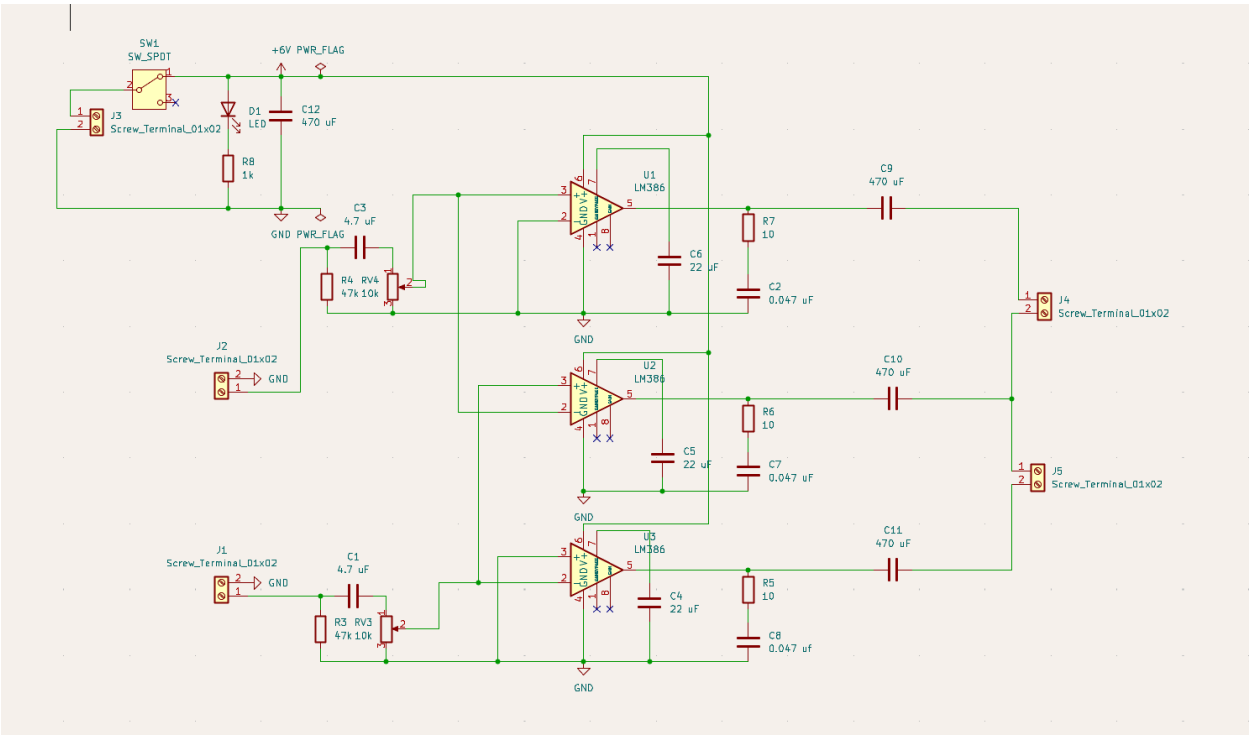
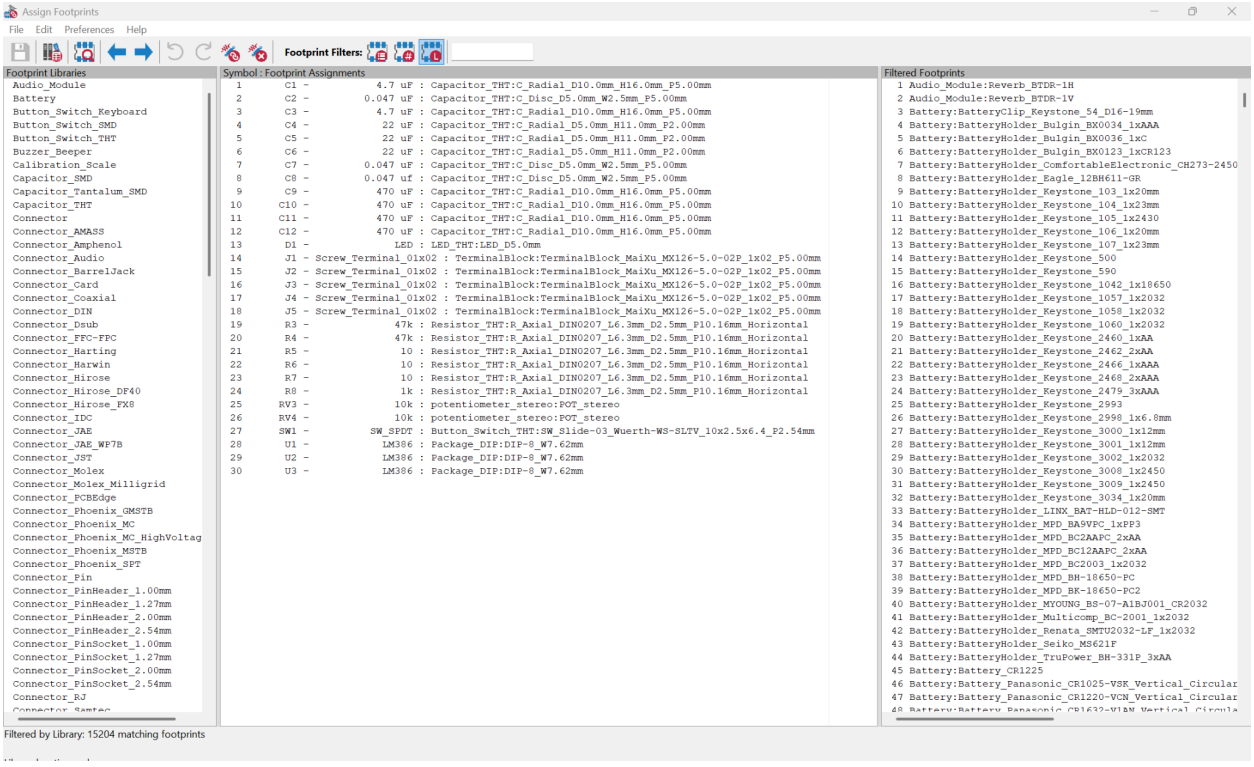
Sohum Tiwary (015641977), Alfredo De La Torre (015634697), Marjery Dilig(017682080)
CMPE110-03
December 3rd, 2025

PCB Design Project Report

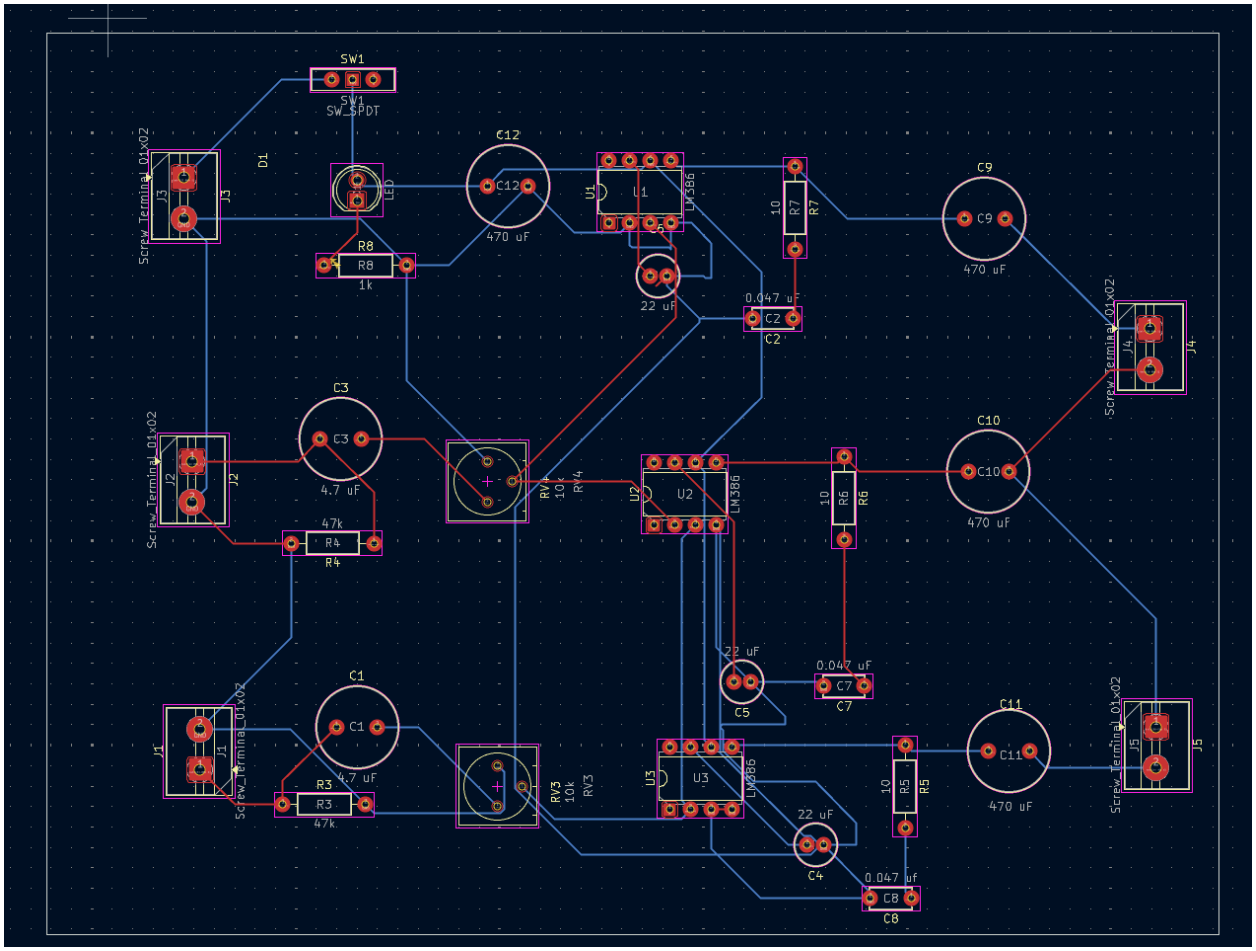
KiCad Schematic:



KiCad Footprint:

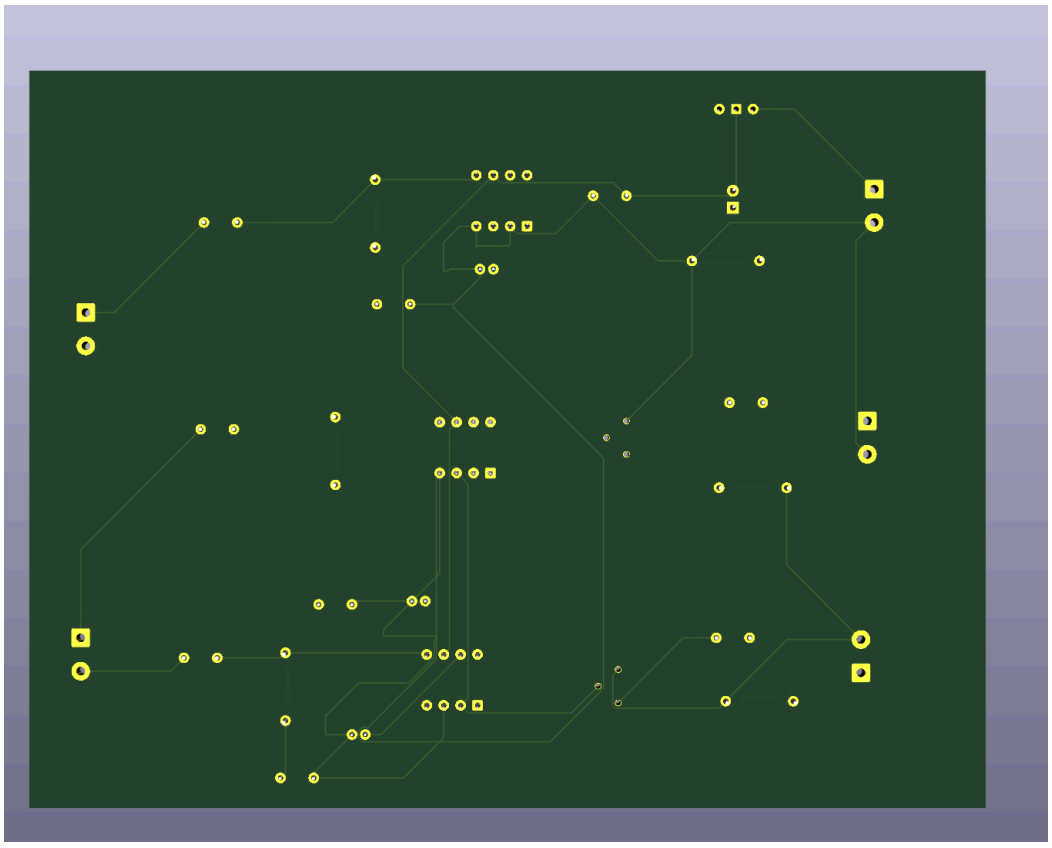


KiCad Routing:



KiCad 3D Model:





Bills of Material:

"Id";"Designator";"Footprint";"Quantity";"Designation";"Supplier and ref";
1;"C6,C5,C4";"C_Radial_D5.0mm_H11.0mm_P2.00mm";3;"22 uF";;;
2;"C10,C12,C9,C11";"C_Radial_D10.0mm_H16.0mm_P5.00mm";4;"470 uF";;;
3;"J5,J2,J1,J4,J3";"TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm";5;"Screw_Terminal_01x02";;;
4;"R3,R4";"R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal";2;"47k";;;
5;"C3,C1";"C_Radial_D10.0mm_H16.0mm_P5.00mm";2;"4.7 uF";;;
6;"SW1";"SW_Slide-03_Wuerth-WS-SLTV_10x2.5x6.4_P2.54mm";1;"SW_SPDT";;;
7;"D1";"LED_D5.0mm";1;"LED";;;
8;"R8";"R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal";1;"1k";;;
9;"RV4,RV3";"POT_stereo";2;"10k";;;
10;"U1,U2,U3";"DIP-8_W7.62mm";3;"LM386";;;
11;"C2,C7";"C_Disc_D5.0mm_W2.5mm_P5.00mm";2;"0.047 uF";;;
12;"R7,R6,R5";"R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal";3;"10";;;
13;"C8";"C_Disc_D5.0mm_W2.5mm_P5.00mm";1;"0.047 uf";;;

Design and Simulation Process

The objective of the project was to design a stereo audio amplifier capable of delivering approximately 2 W per channel with a load (typically 4-8 ohms). The circuit was initially constructed in LTSpice using ideal and manufacturer-provided component models. After the LTSpice simulations validated the design, we moved forward with PCB development.

The next step was creating the schematic on KiCad, assigning the power symbols, alongside with nets being named for clarity. After the schematic was created, the ERC (Electrical Rules Check) was applied to ensure: no unconnected pins, appropriate power flags, correct net labeling, and consistent pin types. On top of that, every component received a unique designation (e.g., R1, C3, U2).

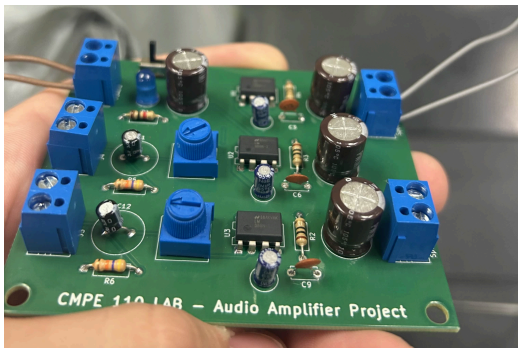
After the schematic, the next step was to assign footprints to each component. The resistors and capacitors were mapped to 0603, 0805, or other chosen SMD/TH packages. ICs were assigned manufacturer-correct footprints. Alongside the footprints, there were 3D models associated with the footprints. The overall board outline was set, with the component height verification being checked in KiCad's 3d viewer.

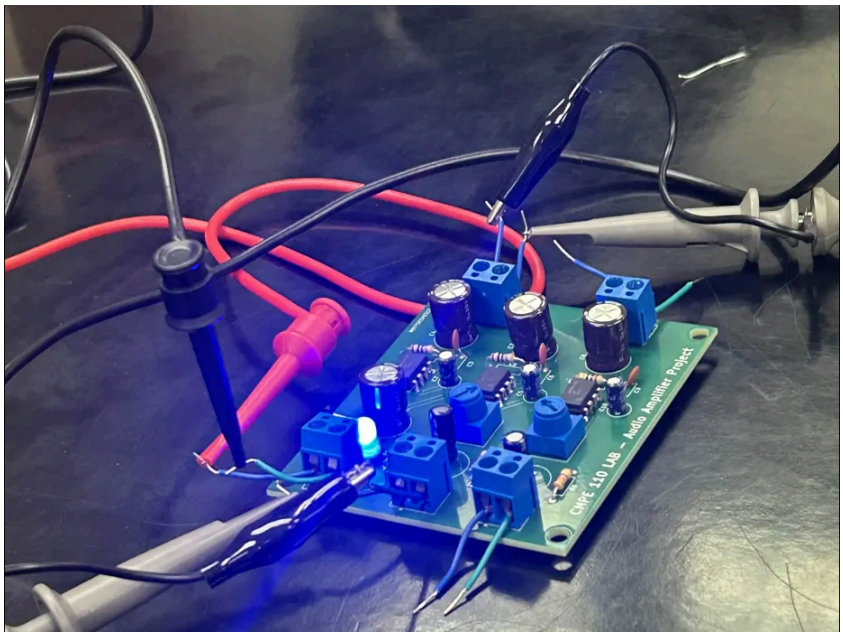
The next step would be the PCB Layout. Design rules are being configured according to the fabrication capabilities, alongside a board outline created to fit the project. The priorities for placements were to shorten signal paths for sensitive audio signals, alongside the separation of

left and right channels to minimize crosstalk. The I/O connectors needed proper positioning. The power traces were routed first with an appropriate width for current requirements. After the layout was done, the DRC (Design Rules Check) ensured all clearances and connections were correct.

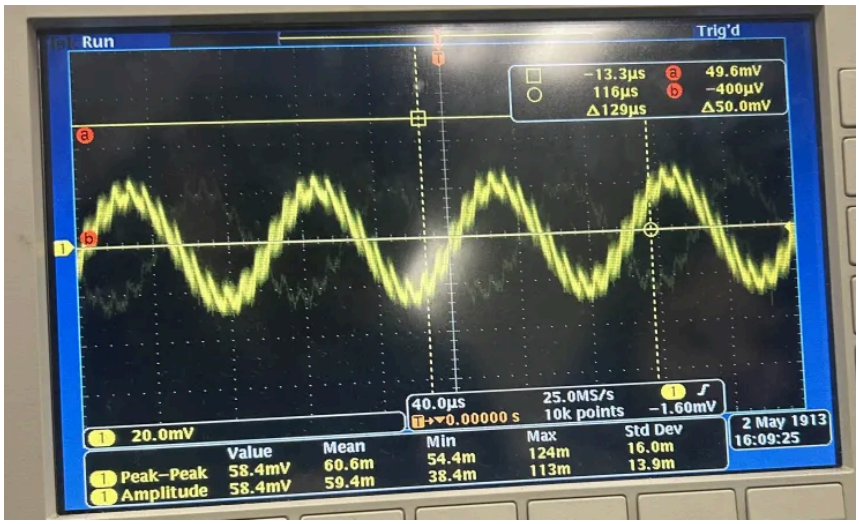
After the layout was finished, the next step was to solder all the components onto the PCB. Soldering was a unique process since there are many dangerous components and factors. Learning the importance of flux, the type of wire when soldering, and the amount of heat set for the solder iron was an imperative experience. However, seeing the metal heat and liquefy was an experience only to be seen from soldering. After the components were soldered onto the PCB, the next step was to test the completed PCB by applying the voltage while observing the speaker output. Hearing the high-frequency noise ensured that the PCB was successful and gave the output needed.

Photos:






Tek Stop
−13.3µs 24.8mV
116µs 400µV
Δ129µs Δ24.4mV
40.0µs 25.0MS/s
T→▼0.00000 s 10k points −1.60mV
1 20.0mV
2 May 1913
16:10:01
Value Mean Min Max Std Dev
1 Peak−Peak 56.0mV 60.0m 54.4m 124m 14.9m
1 Amplitude 56.0mV 59.0m 38.4m 113m 12.9m


Run
Trig'd
−13.3µs 49.6mV
116µs −400µV
Δ129µs Δ50.0mV
40.0µs 25.0MS/s
T→▼0.00000 s 10k points −1.60mV
1 20.0mV
2 May 1913
16:09:25
Value Mean Min Max Std Dev
1 Peak−Peak 58.4mV 60.6m 54.4m 124m 16.0m
1 Amplitude 58.4mV 59.4m 38.4m 113m 13.9m

